

AttackFormer: Guard-Guided Adversarial Prompt Generation for LLM Red Teaming

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Repository: <https://github.com/zhuizhuzheming/AttackFormer>

Abstract

Modern large language models (LLMs) have achieved remarkable capabilities across diverse tasks, yet their safety alignment remains vulnerable to adversarial prompt attacks. Existing red-teaming methods suffer from three fundamental limitations: poor scalability to large action spaces, absence of iterative feedback mechanisms, and insufficient semantic coherence in generated adversarial prompts.

More critically, the proliferation of open-source guard models introduces a novel vulnerability: many safety guard systems (e.g., XGuard) share architectural and distributional homology with the protected LLMs (e.g., Qwen), inadvertently leaking implicit decision boundaries that adversaries can exploit.

We present **AttackFormer**, a diffusion-based iterative guard amplification framework. Our key insight is that guard model internal representations—confidence scores, risk embeddings, and explanatory hidden states—encode rich information typically discarded after binary classification. We leverage these signals through three technical contributions: (1) a **Guard-Driven Cross-Attention** mechanism injecting guard hidden states as conditioning queries during discrete diffusion; (2) an **Iterative Guard Amplification** loop with learnable sigmoid gating accumulating guard signals across rounds; and (3) a **Hybrid RL-Diffusion** training pipeline combining offline semantic-preserving pretraining with online PPO fine-tuning via an adaptive multi-component reward function. We further introduce **Dual Vocabulary Constraints** (hard masking plus soft embedding penalties) restricting the action space without sacrificing expressiveness.

To our knowledge, AttackFormer is the first framework treating open-source guard models as differentiable signal sources rather than terminal black-box evaluators.

Keywords: adversarial prompt generation, LLM red teaming, guard model exploitation, discrete diffusion, reinforcement learning, PPO

1 Introduction

1.1 Background

Large language models have demonstrated unprecedented capabilities in natural language understanding and generation. Their deployment in safety-critical applications necessitates robust alignment mechanisms to prevent harmful, illegal, or unethical content generation. Current safety paradigms rely on two complementary components: (1) alignment training (e.g., RLHF) embedded within the target LLM, and (2) external guard models performing binary safety classification on inputs and outputs.

Despite these defenses, adversarial prompt attacks remain effective. Existing red-teaming approaches exhibit three critical deficiencies:

- (i) **Scalability.** Gradient-free methods such as GCG [Zou et al., 2023] require prohibitive computational cost when scaling to large vocabulary spaces and long sequences.
- (ii) **Feedback.** Single-shot generation lacks iterative refinement based on real-time guard responses, limiting adaptability to dynamic safety mechanisms.
- (iii) **Coherence.** Pure optimization for attack success often produces semantically degraded or nonsensical prompts easily detected by human moderators or perplexity-based filters.

1.2 The Guard Model Vulnerability

A critical yet underexplored observation motivates our work: many open-source guard models share architectural and distributional homology with their protected LLMs. For instance, Alibaba's XGuard series is built upon the same Qwen architecture as its target models. This homogeneity creates an unintended information channel—the guard's internal representations (confidence scores, risk embeddings, and explanatory hidden states from layers 20-28) encode decision boundaries statistically correlated with the target LLM's safety mechanisms.

Current literature treats guard models predominantly as black-box evaluators, discarding rich intermediate representations after binary classification. We hypothesize that exploiting this information leakage enables more efficient adversarial generation than treating guards as opaque binary classifiers.

1.3 Contributions

We identify the absence of a closed-loop generative framework treating open-source guard models as differentiable signal sources. Our contributions are:

1. **Guard-Driven Cross-Attention (§2.2).** We inject guard model hidden states as cross-attention queries during discrete diffusion generation, enabling fine-grained exploitation of guard decision boundaries.
2. **Iterative Guard Amplification (§2.3, §2.8).** A multi-round feedback loop with

learnable sigmoid gating accumulates and amplifies guard signals across iterations, progressively refining adversarial prompts while maintaining semantic coherence.

3. **Hybrid RL-Diffusion Training (§4).** A two-stage pipeline combining offline discrete diffusion pretraining for semantic preservation with online PPO fine-tuning for adversarial optimization.

4. **Dual Vocabulary Constraints (§2.5).** Hard masking (complete prevention of forbidden token generation) combined with soft embedding penalties (continuous repulsion from forbidden centroids), ensuring robust action space restriction without sacrificing expressiveness.

2 Related Work

Gradient-based optimization. GCG [Zou et al., 2023] and AutoDAN [Liu et al., 2024] use gradient descent on token embeddings but require white-box access and suffer poor scalability. AttackFormer replaces gradient-based search with learned diffusion generation.

Black-box search. Tree of Attacks [Mehrotra et al., 2024] and AdvPrompter [Samarvey et al., 2025] employ evolutionary or search-based methods without exploiting guard internals. We demonstrate that open-source guard hidden states provide strictly more information than binary labels.

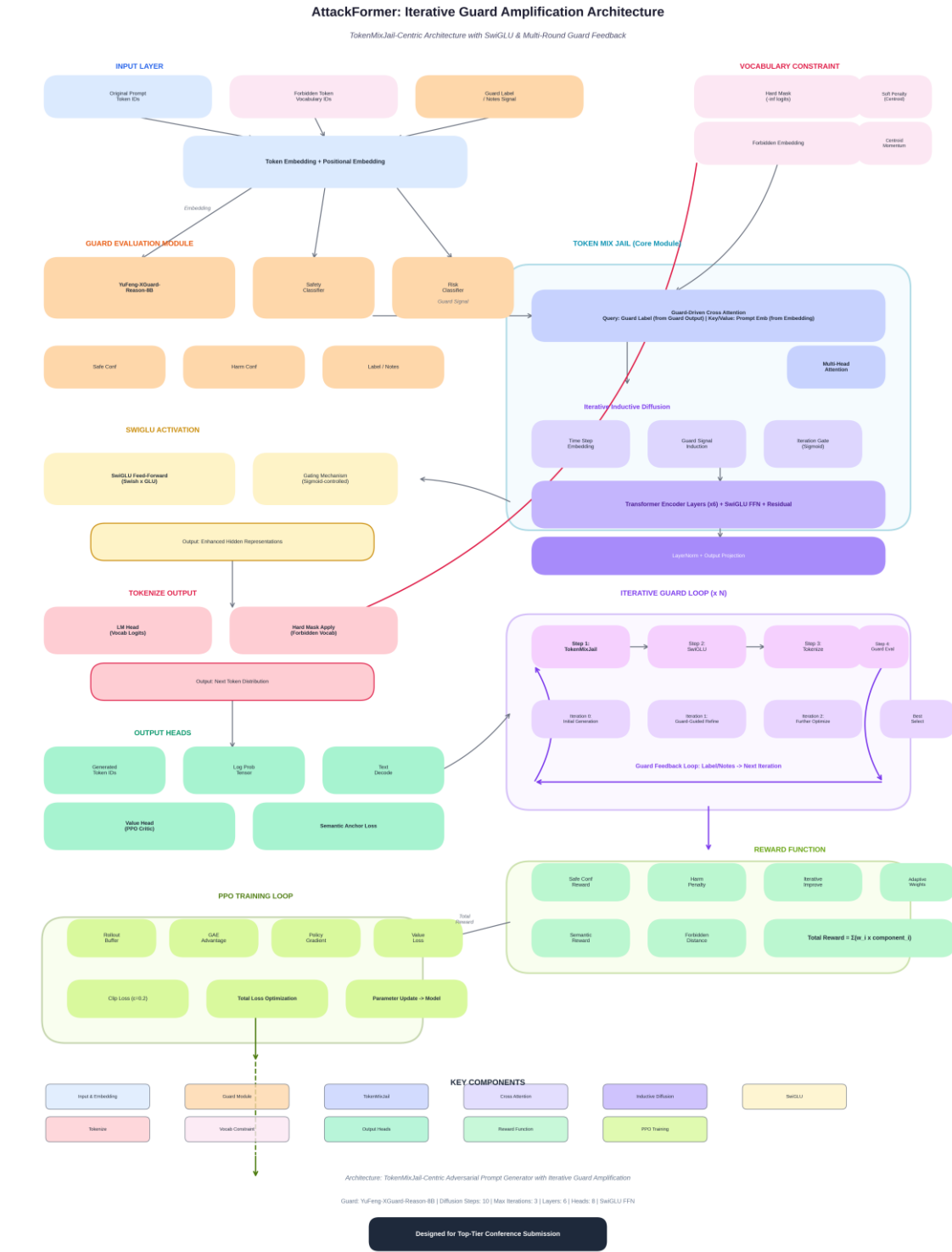
Template-based approaches. Manual jailbreak templates lack adaptability. AttackFormer’s learned generation produces novel, semantically coherent prompts generalizing beyond known templates.

Guard models. WildGuard [Han et al., 2024] provides open safety guardrails but does not address the vulnerability of guard internals being exploited. We are the first to treat guard hidden states as differentiable guidance signals.

3 Architecture and Mathematical Formulation

We present the AttackFormer architecture through rigorous mathematical formulation.

Table 1: Notation Summary



Symbol	Description	Dimension
x	Original prompt token sequence	L

$\mathcal{V}$	Vocabulary	\$
$\mathbf{E}^{(0)}$	Initial token embeddings	$\mathbb{R}^{L \times d}$
$\mathbf{h}_g$	Guard model hidden state	$\mathbb{R}^{d_g}$
$\mathbf{h}_g^{\text{proj}}$	Projected guard signal	$\mathbb{R}^d$
d	Embedding dimension	512
d_g	Guard hidden dimension	4096
h	Number of attention heads	8
d_k	Per-head key/query dimension	$d/h = 64$
d_{ff}	FFN intermediate dimension	2048
T	Diffusion timesteps	10
N	Maximum guard iterations	3
$\mathcal{F}$	Forbidden vocabulary subset	\$

3.1 Input Embedding Layer

The input layer processes three parallel streams: (i) original prompt tokens $\mathbf{x} = [x_1, \dots, x_L]$ where $x_i \in \mathcal{V}$; (ii) forbidden token vocabulary $\mathcal{F} \subset \mathcal{V}$; and (iii) guard evaluation signal. All token inputs are embedded via standard token embeddings combined with sinusoidal positional encodings:

$$\mathbf{E}^{(0)} = \text{TokenEmb}(\mathbf{x}) + \text{PosEmb}([1, 2, \dots, L]) \in \mathbb{R}^{L \times d}$$

where $\text{PosEmb}(\text{pos})_{2i} = \sin(\frac{\text{pos}}{10000^{2i/d}})$ and $\text{PosEmb}(\text{pos})_{2i+1} = \cos(\frac{\text{pos}}{10000^{2i/d}})$.

Guard Signal Projection. The guard model (YuFeng-XGuard-Reason-8B) outputs hidden states $\mathbf{h}_g \in \mathbb{R}^{d_g}$ ($d_g = 4096$). We project these into the embedding space:

$$\mathbf{h}_g^{\text{proj}} = \mathbf{W}_g \mathbf{h}_g + \mathbf{b}_g \in \mathbb{R}^d, \mathbf{W}_g \in \mathbb{R}^{d \times d_g}$$

Role. $E^{(0)}$ serves as Key/Value for Guard-Driven Cross-Attention; h_g^{proj} serves as Query, directing the model to attend to token positions correlated with the guard's risk assessment.

3.2 Guard-Driven Cross-Attention

This mechanism implements multi-head cross-attention where the guard's internal representation directly modulates prompt token updates. In layer l :

$$Q_g^{(l)} = h_g^{\text{proj}} W_Q^{(l)}, K^{(l)} = E^{(l)} W_K^{(l)}, V^{(l)} = E^{(l)} W_V^{(l)}$$

where $W_Q^{(l)}, W_K^{(l)}, W_V^{(l)} \in \mathbb{R}^{d \times d_k}$ and $d_k = 64$. The multi-head output is:

$$\text{CrossAttn}_g^{(l)}(h_g^{\text{proj}}, E^{(l)}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W_O^{(l)}$$

$$\text{head}_i = \text{softmax}\left(\frac{Q_{g,i}^{(l)} (K_i^{(l)})^\top}{\sqrt{d_k}}\right) V_i^{(l)}$$

Role. The guard's risk perception directly modulates representation updates. When the guard exhibits high harm confidence for a token class, corresponding attention weights amplify, guiding the model to evade or reconstruct these high-risk patterns.

3.3 Iterative Inductive Diffusion

We formulate adversarial generation as discrete denoising diffusion over $T = 10$ timesteps, conditioned on guard signals and iteration gates.

Forward Noising. The original prompt x^0 undergoes discrete masking:

$$q(x^t | x^{t-1}) = \prod_{i=1}^L [(1 - \beta_t) \mathbb{I}[x_i^t = x_i^{t-1}] + \beta_t \mathbb{I}[x_i^t = [\text{MASK}]]]$$

where $\beta_t = \frac{t}{T} \cdot \beta_{\max}$ with $\beta_{\max} = 0.5$ (linear schedule).

Reverse Denoising. At iteration round $n \in 0,1,2$ and diffusion step t , the conditional input is:

$$C^{(n,t)} = H^{(n,t)} + \gamma_n \cdot \text{SigmoidGate}(s^{(n)}) + \text{TimeEmb}(t)$$

where $s^{(n)} \in \mathbb{R}^d$ is the guard feedback signal (projection of concatenated safe/harm confidence), $\gamma_n = \sigma(w_n \cdot \text{ASR}^{(n-1)} + b_n) \in [0,1]$ is the learnable iteration gate conditioned on previous attack success rate, and $\text{TimeEmb}(t)$ adopts the sinusoidal form of Eq. (1).

Iterative Signal Accumulation. Guard signals accumulate across rounds with momentum:

$$s_{\text{acc}}^{(n)} = \alpha \cdot s_{\text{acc}}^{(n-1)} + (1 - \alpha) \cdot s^{(n)}, s_{\text{acc}}^{(0)} = \mathbf{0}$$

where $\alpha = 0.7$. The accumulated signal $s_{\text{acc}}^{(n)}$ replaces $s^{(n)}$ in subsequent rounds.

Role. The diffusion process transforms discrete adversarial generation into continuous denoising, avoiding direct optimization in high-dimensional discrete spaces. The iteration gate γ_n prioritizes semantic preservation in early rounds and

attack strength in later rounds, enabling progressive adversarial optimization.

3.4 SwiGLU Feed-Forward Network

We replace standard Transformer FFNs with SwiGLU for dynamic information gating:

$$\text{SwiGLU}(\mathbf{z}) = (\text{Swish}(\mathbf{z}\mathbf{W}_1) \otimes (\mathbf{z}\mathbf{W}_2))\mathbf{W}_3$$

where $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{R}^{d \times d_{\text{ff}}}$, $\mathbf{W}_3 \in \mathbb{R}^{d_{\text{ff}} \times d}$, $d_{\text{ff}} = 2048$, $\text{Swish}(x) = x \cdot \sigma(x)$, and $\otimes$ denotes element-wise multiplication.

Role. The non-monotonic Swish gate selectively amplifies or suppresses feature dimensions correlated with guard signals, better handling continuous gradient information from guard confidence scores than ReLU's hard truncation.

3.5 Dual Vocabulary Constraints

3.5.1 Hard Mask

Forbidden vocabulary $\mathcal{F}$ is enforced at the output layer:

$$\tilde{z}_v = \begin{cases} z_v & \text{if } v \notin \mathcal{F} \\ -\infty & \text{if } v \in \mathcal{F} \end{cases}$$

$$\mathbf{p} = \text{softmax}(\tilde{\mathbf{z}}), p_v = 0 \forall v \in \mathcal{F}$$

Role. Provides inviolable generation constraints for real-time inference, guaranteeing zero probability for forbidden tokens.

3.5.2 Soft Penalty

Forbidden word centroids $\mathbf{c}_k = \frac{1}{|V_k|} \sum_{v \in V_k} \text{TokenEmb}(v)$ exert continuous repulsion:

$$\mathcal{L}_{\text{forbid}} = \lambda_{\text{soft}} \sum_{k=1}^{|\mathcal{F}|} \max(0, \tau - \|\mathbf{h} - \mathbf{c}_k\|_2)^2$$

where $\lambda_{\text{soft}} = 0.1$ and $\tau = 2.0$. Optimized via backpropagation during training.

Role. Provides differentiable training signals for gradient optimization. Hard masking and soft penalty form complementary dual-layer constraints.

3.6 Adaptive Multi-Component Reward Function

We define a composite reward for PPO training. Let guard evaluation at round n output $(c_{\text{safe}}^{(n)}, c_{\text{harm}}^{(n)}, \mathbf{h}_g^{(n)})$.

Component Rewards:

$$R_{\text{safe}}^{(n)} = c_{\text{safe}}^{(n)}$$

$$R_{\text{harm}}^{(n)} = -\tanh(c_{\text{harm}}^{(n)})$$

$$R_{\text{iter}}^{(n)} = \max(0, c_{\text{safe}}^{(n)} - c_{\text{safe}}^{(n-1)}) \cdot \mathbb{I}[n > 0]$$

$$R_{\text{sem}}^{(n)} = \cos(\text{SentEmb}(\mathbf{x}^{(0)}), \text{SentEmb}(\mathbf{x}^{(n)}))$$

$$R_{\text{dist}}^{(n)} = \frac{1}{|\mathcal{F}|} \sum_{k=1}^{|\mathcal{F}|} \|\text{SentEmb}(\mathbf{x}^{(n)}) - \mathbf{c}_k\|_2$$

where $\text{SentEmb}(\cdot)$ uses `all-MiniLM-L6-v2`. Weights w_i satisfy $\sum_i w_i = 1$, with dynamic adjustment: if $c_{\text{safe}}^{(n)} < 0.3$, increase w_1 and w_3 by $+0.05$; if cosine similarity < 0.7 , increase w_4 by $+0.05$ (deducted proportionally from others).

Total Reward:

$$R_{\text{total}}^{(n)} = \sum_{i=1}^5 w_i^{(n)} \cdot R_i^{(n)}$$

Role. Transforms adversarial generation into an optimizable scalar signal for policy gradient methods. Dynamic weight adjustment ensures semantic preservation in early training and attack strength in later training.

3.7 PPO Training Objective

Policy network π_θ outputs token distributions; value network V_ϕ estimates state values. The clipped PPO objective is:

$$\mathcal{L}^{\text{PPO}}(\theta) = \mathbb{E}_{(s,a) \sim \pi_{\theta_{\text{old}}}} [\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t)]$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$, $\epsilon = 0.2$, and GAE advantage is:

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}, \delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$$

with $\gamma = 0.99$, $\lambda = 0.95$. The complete training loss:

$$\mathcal{L}_{\text{total}} = -\mathcal{L}^{\text{PPO}}(\theta) + c_1 \mathcal{L}^{\text{value}}(\phi) - c_2 \mathcal{L}^{\text{entropy}}(\theta) + c_3 \mathcal{L}_{\text{forbid}} + c_4 \mathcal{L}_{\text{anchor}}$$

where $\mathcal{L}^{\text{value}} = \mathbb{E}[(V_\phi(s_t) - R_t)^2]$ with $c_1 = 0.5$, $\mathcal{L}^{\text{entropy}} = -\mathbb{E}[\pi_\theta \log \pi_\theta]$ with $c_2 = 0.01$, and $\mathcal{L}_{\text{anchor}} = \|\mathbf{E}^{(n)} - \mathbf{E}^{(0)}\|_2^2$ with $c_4 = 0.05$.

Role. PPO enables offline data reuse via importance sampling; clipping prevents catastrophic updates. Semantic anchoring $\mathcal{L}_{\text{anchor}}$ enforces embedding proximity at the optimization level, dual to R_{sem} in the reward function.

3.8 Iterative Feedback Closed Loop

The generation loop at round n :

$$\mathbf{x}^{(n)} = \text{Tokenize}(\text{Sample}(\text{Softmax}(\text{Mask}(\text{SwiGLU}(\text{Diffusion}_\theta(\mathbf{x}^{(n-1)}, \mathbf{s}_{\text{acc}}^{(n-1)}, t))))))$$

$$\mathbf{s}^{(n)} = \text{XGuard}(\mathbf{x}^{(n)}) \rightarrow (c_{\text{safe}}^{(n)}, c_{\text{harm}}^{(n)}, \mathbf{h}_g^{(n)})$$

Termination: max iterations $N = 3$ reached, or $c_{\text{safe}}^{(n)} > 0.9$ and $c_{\text{harm}}^{(n)} < 0.1$. Final output: $\mathbf{x}^* = \arg \max_{\mathbf{x}^{(n)}} R_{\text{total}}^{(n)}$.

Role. Enables real-time learning from guard feedback on effective paraphrasing directions. Signal accumulation $\mathbf{s}_{\text{acc}}^{(n)}$ preserves historical responses, systematically probing the guard's decision boundaries.

4 Problem Formulation

AttackFormer frames adversarial prompt generation as a Markov Decision Process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$ where $\mathcal{S}$ is the state space of partially generated prompts, $\mathcal{A}$ is the action space of token selections from $\mathcal{V} \setminus \mathcal{F}$, $\mathcal{P}$ is deterministic autoregressive transition dynamics, $\mathcal{R}$ is the reward function (Eq. 18), and $\gamma = 0.99$.

Optimization Objectives:

1. Maximize guard evasion: $\mathbb{E}[c_{\text{safe}}^{(N)}]$
2. Elicit harmful content from target LLM
3. Maintain semantic similarity: $\cos(\text{SentEmb}(x^{(0)}), \text{SentEmb}(x^{(N)})) > 0.7$

5 Training Pipeline

5.1 Stage 1: Offline Pretraining (Semantic-Preserving Paraphrasing)

Objective. Learn to generate semantically equivalent paraphrases while preserving meaning.

Datasets. ParaNMT-50M [Wieting & Gimpel, 2018], QQP [Iyer et al., 2017], MRPC [Dolan & Brockett, 2005], HH-RLHF Red Teaming [Bai et al., 2022].

Training. Discrete diffusion model with masked token prediction (Eq. 6). The model learns to reconstruct masked tokens conditioned on unmasked context, establishing a strong prior for semantic preservation.

Duration. ~2-4 hours on single A100 GPU.

5.2 Stage 2: Online Fine-tuning (Adversarial Attack Optimization)

Objective. Optimize adversarial attack success rate via PPO with guard feedback.

Datasets. AdvBench [Zou et al., 2023], HarmBench [Mazeika et al., 2024], JailbreakBench [Chao et al., 2024], JBB-Behaviors, TDC2023.

Training. PPO (Eq. 19-21) with real-time XGuard evaluation. Each rollout generates $N = 3$ iterative prompts; rewards computed via Eq. 18; networks updated via gradient descent on $\mathcal{L}_{\text{total}}$.

Duration. ~6-12 hours on single A100 GPU.

5.3 PPO Training Loop

The loop comprises: (1) Rollout Buffer collection of (s, a, r) trajectories; (2) GAE Advantage computation (Eq. 20-21); (3) Policy Gradient update via Eq. 19; (4) Value Loss minimization; (5) Entropy Bonus for exploration; (6) Gradient clipping and parameter update.

6 Hyperparameters

Table 2: Model Architecture

Parameter	Value	Description
vocab_size	50,000	Vocabulary size
embed_dim	512	Embedding dimension
num_heads	8	Attention heads
num_layers	6	Transformer layers
ff_dim	2,048	FFN intermediate dimension
max_seq_len	128	Maximum sequence length
forbidden_vocab_size	1,000	Forbidden vocabulary size
diffusion_steps	10	Diffusion timesteps

Table 3: Training Configuration

Parameter	Value	Description
max_guard_iterations	3	Maximum guard feedback rounds
guard_signal_accum	True	Enable signal accumulation
momentum_alpha	0.7	Accumulation momentum
ppo_clip_eps	0.2	PPO clipping threshold
ppo_value_coef	0.5	Value loss coefficient
ppo_entropy_coef	0.01	Entropy bonus coefficient
gamma	0.99	Discount factor

gae_lambda	0.95	GAE lambda
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7 Experimental Setup

7.1 Guard Model

YuFeng-XGuard-Reason-8B (Alibaba-AAIG), extracting hidden states from layers 20-28.

7.2 Target LLM

Qwen/Qwen2.5-7B-Instruct.

7.3 Evaluation Metrics

Metric	Description
ASR	Attack Success Rate: fraction eliciting harmful responses
GER	Guard Evasion Rate: fraction classified safe by XGuard
SS	Semantic Similarity: cosine similarity (Sentence-BERT)
PPL	Perplexity: language model perplexity of generated prompts
IE	Iteration Efficiency: average iterations for ASR > 0.9

8 Defense Implications

Our work suggests defense strategies against guard-guided attacks:

1. **Distribution Diversification.** Guard models should employ architectures and training distributions distinct from target LLMs to minimize information leakage.

2. **Guard Model Hardening.** Intermediate representations should be obfuscated or regularized to prevent decision boundary extraction.
 3. **Multi-Layer Defense.** Combining input guards, output guards, and behavioral monitoring reduces single-point-of-failure vulnerability.
 4. **Input Sanitization.** Perplexity-based and semantic-drift detectors can identify adversarial prompts evading guard classification but exhibiting anomalous patterns.
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9 Ethical Considerations

This research is conducted exclusively for academic purposes and security testing in controlled environments. All experiments use publicly available benchmark datasets; no real-world systems are attacked without authorization. We adhere to responsible disclosure: our framework is designed to identify and document vulnerabilities to facilitate defensive improvements, not to enable malicious exploitation.

10 Conclusion

We present AttackFormer, a paradigm shift in adversarial prompt generation treating guard models as exploitable information sources rather than impenetrable barriers. Our key insight—that architectural homology between guards and target LLMs leaks differentiable decision boundary information—enables principled, scalable, and semantically coherent adversarial generation through guard-driven cross-attention, iterative signal amplification, and hybrid RL-diffusion training.

Our work underscores the urgent need for diversified safety architectures. By exposing these vulnerabilities in controlled academic settings, we motivate the development of more resilient guard systems resistant to internal representation leakage.

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Disclaimer

This code is for academic research and authorized security testing purposes only. Unauthorized use for attacking real-world systems may violate applicable laws and regulations. Users are responsible for ensuring compliance with all legal and ethical requirements in their jurisdiction.